

MEDOFLUCON®

COMPOSITION

Each "MEDOFLUCON" capsule 150mg, contains 150mg of fluconazole

INDICATIONS

Therapeutic Indications

Therapy may be instituted before the results of the cultures and the other laboratory studies are known; however, once these results become available, anti-infective therapy should be adjusted accordingly.

1. Cryptococcosis, including cryptococcal meningitis and infections of other sites (e.g: pulmonary, cutaneous). Normal hosts and patients with AIDS, organ transplant or other causes of immunosuppression may be treated. Fluconazole can be used as maintenance therapy to prevent relapse of cryptococcal disease in patients with AIDS.
2. Systemic candidiasis, including candidemia, disseminated candidiasis and other forms of invasive candidal infections. These include infections of the peritoneum, endocardium, eye and pulmonary and urinary tracts. Patients with malignancy, in intensive care units, receiving cytotoxic or immunosuppressive therapy, or with other factors predisposing to candidal infection may be treated.
3. Mucosal candidiasis. These include oropharyngeal, esophageal, non-invasive bronchopulmonary infections, candiduria, mucocutaneous and chronic oral atrophic candidiasis (denture sore mouth). Normal hosts and patients with compromised immune function may be treated. Prevention of relapse of oropharyngeal candidiasis in patients with AIDS.
4. Genital candidiasis. Vaginal candidiasis, acute or recurrent; and prophylaxis to reduce the incidence of recurrent vaginal candidiasis (3 or more episodes a year). Candidal balanitis.
5. Prevention of fungal infections in patients with malignancy who are predisposed to such infections as a result of cytotoxic chemotherapy or radiotherapy.
6. Dermatomycosis including tinea pedis, tinea corporis, tinea cruris, tinea versicolor, and dermal candida infections.

DOSAGE AND METHOD OF ADMINISTRATION

Posology and Method of Administration

The daily dose of fluconazole should be based on the nature and severity of the fungal infection. Most cases of vaginal candidiasis respond to single dose therapy. Therapy for those types of infections requiring multiple dose treatment should be continued until clinical parameters or laboratory tests indicate that active fungal infection has subsided. An adequate period of treatment may lead to recurrence of active infections. Patients with AIDS and cryptococcal meningitis or recurrent oropharyngeal candidiasis usually require maintenance therapy to prevent relapse.

In adults

1. For cryptococcal meningitis and cryptococcal infections at other sites, the usual dose is 400 mg on the first day followed by 200 mg – 400 mg once daily. Duration of treatment for cryptococcal infections will depend on the clinical and mycological response, but is usually at least 6-8 weeks for cryptococcal meningitis. For the prevention of relapse of cryptococcal meningitis in patients with AIDS, after the patient receives a full course of primary therapy, fluconazole may be administered indefinitely at a daily dose of 200 mg.
2. For candidemia, disseminated candidiasis and other invasive candidal infections, the usual dose is 400 mg on the first day followed by 200 mg daily. Depending on the clinical response, the dose may be increased to 400mg daily. Duration of treatment is based upon the clinical response.
3. For oropharyngeal candidiasis, the usual dose is 50 to 100 mg once daily for 7-14 days. If necessary treatment can be continued for longer periods in patients with severely compromised immune function. For atrophic oral candidiasis associated with dentures, the usual dose is 50 mg once daily for 14 days administered concurrently with local antiseptic measures to the denture. For other candidal infections of mucosa except genital candidiasis (see below), (eg: esophagitis, noninvasive bronchopulmonary infections, candiduria, mucocutaneous candidiasis, etc), the usual effective dose is 50 to 100 mg daily, given for 14-30 days. For the prevention of relapse of oropharyngeal candidiasis in patients with AIDS, after the patient receives a full course of primary therapy, fluconazole may be administered at a 150 mg once weekly dose.
4. For the treatment of vaginal candidiasis, fluconazole 150 mg should be administered as a single oral dose. To reduce the incidence of recurrent vaginal candidiasis, a 150 mg once monthly dose may be used. The duration of therapy should be individualized but ranges from 4-12 months. Some patients may require more frequent dosing. For candida balanitis, fluconazole 150 mg should be administered as a single oral dose.
5. The recommended fluconazole dosage for the prevention of candidiasis is 50 to 400 mg once daily, based on the patient's risk for developing fungal infection. For patients at high risk of systemic infection e.g patients who are anticipated to have profound or prolonged neutropenia, the recommended daily dose is 400 mg once daily. Fluconazole administration should start several days before the anticipated onset of neutropenia and continue for 7 days after the neutrophil count rises above 1000 cells per mm³.

6. For dermal infections including tinea pedis, corporis, cruris and candida infections the recommended dosage is 150 mg once weekly or 50 mg once daily. Duration of treatment is normally 2 to 4 weeks but tinea pedis may require treatment for up to 6 weeks. For tinea versicolor, the recommended dose is 300 mg once weekly for 2 weeks, a third weekly dose of 300 mg may be needed in some patients, whereas, in some patients, a single dose of 300- 400 mg may be sufficient. An alternate dosing regimen is 50 mg once daily for 2 to 4 weeks.

****In children***

As with similar infections in adults, the duration of treatment is based on the clinical and mycological response. The maximum adult daily dosage should not be exceeded in children. Fluconazole is administered as a single dose each day. The recommended dosage of Fluconazole for mucosal candidiasis is 3mg/kg daily. A loading dose of 6mg/kg may be used on the first day to achieve steady state levels more rapidly.

For the treatment of systemic candidiasis and cryptococcal infections, the recommended dosage is 6-12 mg/kg daily depending on the severity of the disease.

For the prevention of fungal infections in immunocompromised patients considered at risk as a consequences of neutropenia following cytotoxic chemotherapy or radiotherapy, the dose should be 3-12 mg/kg daily, depending on the extent and duration of the induced neutropenia (see adult dosing).

For children with impaired renal function the daily dose should be reduced in accordance with the guidelines given for adults, dependent on the degree of the renal impairment.

****Children below 4 weeks of age***

Neonates excrete Fluconazole slowly. In the first two weeks of life, the same mg/kg dosing as in older should be used but administered every 72 hours. During weeks 2-4 of life, the same dose should be given every 48 hours.

Use in elderly

Where there is no evidence of renal impairment, normal dosage recommendations should be adopted. For patients with renal impairment (creatinine clearance <50ml/min) the dosag schedule should be adjusted as described below.

In patients with renal impairment

Fluconazole is predominantly excreted in the urine as unchanged drug. No adjustments in single-dose therapy are necessary. In patients with impaired renal function who will receive multiple doses of Fluconazole, an initial loading dose of 50 mg to 400 mg should be given. After the loading dose, the daily dose (according to indication) should be based on the following table:

Creatinine Clearance (ml/min)	Percent of Recommended Dose
>50	100%
11-50	50%
Patients receiving regular hemodialysis	One dose after each dialysis session

These are suggested dose adjustments based on pharmacokinetics following administration of multiple doses. Further adjustment may be needed depending upon clinical condition.

CAPSULE IS NOT RECOMMENDED FOR CHILDREN

ROUTE OF ADMINISTRATION: Oral

CONTRA-INDICATIONS

Patients with known hypersensitivity to fluconazole or related azole compounds.

SPECIAL WARNINGS AND PRECAUTIONS FOR USE

Some patients, especially those with serious underlying disease, such as AIDS or cancer, treated with fluconazole have been observed to have abnormalities of renal, hepatic, haematological and other biochemical function tests. Clinical significance and causality is uncertain.

In very rare cases, patients with severe underlying disease and treated with fluconazole, who died were found to have hepatic necrosis. All were on concomitant multiple medication, some of which were known to be hepatotoxic, or had underlying disease that could have caused hepatic necrosis. As a causal relationship with fluconazole cannot be excluded, in patients with a significant rise in liver enzymes the risk benefit of continued fluconazole therapy should be evaluated.

Patients have, in rare cases, developed exfoliative cutaneous reactions, including Stevens-Johnson Syndrome and toxic epidermal necrolysis, during fluconazole therapy. Development of such severe cutaneous reactions is more likely in AIDS patients.

If rash develops during superficial fungal infection therapy that is attributable to fluconazole, therapy should be discontinued. If it develops in patients being treated for invasive/systemic infection, they should be monitored closely and therapy discontinued if bullous lesions or erythema multiforme develops.

INTERACTIONS WITH OTHER MEDICAMENTS AND OTHER FORMS OF INTERACTION

Anticoagulants: In an interaction study, fluconazole increased the prothrombin time after warfarin administration in healthy males. Though the magnitude of change was small (12%), careful monitoring of prothrombin time in patients receiving coumarin-type anticoagulants is recommended.

Sulfonylureas: Fluconazole has been shown to prolong the serum half-life of concomitantly administered oral sulfonylureas (chlorpropamide, glibenclamide, glipizide and tolbutamide) in healthy volunteers. Fluconazole and oral sulfonylureas may be coadministered to diabetic patients, but the possibility of a hypoglycemic episode should be borne in mind.

Hydrochlorothiazide: In a kinetic interaction study, coadministration of multiple-dose hydrochlorothiazide to healthy volunteers receiving fluconazole increased plasma concentrations of fluconazole by 40%. An effect of this magnitude should not necessitate a change in the fluconazole dose regimen in subjects receiving concomitant diuretics, although the prescriber should bear it in mind.

Phenytoin: Concomitant administration of fluconazole and phenytoin may increase the levels of phenytoin to clinically significant degree. If it is necessary to administer both drugs concomitantly, phenytoin levels should be monitored and phenytoin dose adjusted to maintain therapeutic levels.

Oral Contraceptive: Two kinetic studies with a combined oral contraceptive have been performed using multiple doses of fluconazole. There were no relevant effects on either hormone level in the 50 mg fluconazole study, while at 200 mg daily, the AUCs ethinyl estradiol and levonorgestrel were increased 40% and 24%, respectively. Thus, multiple dose use of fluconazole at these doses is unlikely to have an effect on the efficacy of the combined oral contraceptive.

Rifampicin: Concomitant administration of fluconazole and rifampicin resulted in 25% decrease in the AUC and 20% shorter half-life of fluconazole. In patients receiving concomitant rifampicin, an increase of the fluconazole dose should be considered.

Cyclosporin: A kinetic study in renal transplant patient found fluconazole 200 mg daily to slowly increase cyclosporine concentration. However, in another multiple dose study with 100mg daily, fluconazole did not affect cyclosporine levels in patients in bone marrow transplant. Cyclosporin plasma concentration monitoring in patient receiving fluconazole is recommended.

Theophylline: In a placebo controlled interaction study, the administration of fluconazole 200 mg for 14 days resulted in an 18% decrease in the mean plasma clearance rate of theophylline. Patients who are receiving high dose theophylline or who are otherwise at increased risk for theophylline toxicity should be observed for signs of theophylline toxicity while receiving fluconazole, and therapy modified appropriately if signs of toxicity develop.

Terfenadine: Because of the occurrence of serious cardiac dysrhythmias secondary to prolongation of the QTc interval in patients receiving azole antifungals in conjunction with terfenadine, interaction studies have been performed. One study at a 200 mg dose of fluconazole failed to demonstrate a prolongation in QTc interval. Another study 400 mg and 800 mg daily dose of fluconazole demonstrated that fluconazole taken in doses of 400 mg per day or greater significantly increases plasma levels of terfenadine when taken concomitantly. The combined use of fluconazole at doses of 400 mg or greater with terfenadine is contraindicated. (See Contraindications). The coadministration of fluconazole at doses lower than 400 mg per day with terfenadine should be carefully monitored.

Cisapride: There have been reports of cardiac events including torsade de pointes in patients to whom fluconazole and cisapride were coadministered.

Rifabutin: There have been reports that an interaction exists when fluconazole is administered concomitantly with rifabutin, leading to increased serum level of rifabutin. There have been reports of uveitis in patient to whom fluconazole and rifabutin were coadministered. Patients receiving rifabutin and fluconazole concomitantly should be carefully monitored.

Tacrolimus: There have been reports that an interaction exists when fluconazole is administered concomitantly with tacrolimus, leading to increased serum levels of tacrolimus. There have been reports of nephrotoxicity in patients to whom fluconazole and tacrolimus were coadministered. Patients receiving tacrolimus and fluconazole concomitantly should be carefully monitored.

Zidovudine: Two kinetic studies resulted in increased levels of zidovudine most likely caused by the decreased conversion of zidovudine to its major metabolite. One study determined zidovudine levels in AIDS or ARC patients before and following fluconazole 200 mg daily for 15 days. There was a significant increase in zidovudine AUC (20%). A second randomized, two-period, two-treatment cross-over study examined zidovudine levels in HIV infected patients. On two occasions, 21 days apart, patients received zidovudine 200 mg every eight hours either with or without fluconazole 400 mg daily for seven days. The AUC of zidovudine significantly increased (74%) during coadministration with fluconazole. Patients receiving this combination should be monitored for the development of zidovudine –related adverse reactions.

The use of fluconazole in patients concurrently taking cisapride, astemizole, rifabutin, tacrolimus or other drugs metabolized by the cytochrome P-450 system may be associated with elevations in serum levels of these drugs. In the absence of definitive information, caution should be used when coadministering fluconazole. Patients should be carefully monitored.

Interaction studies have shown that when oral fluconazole is coadministered with food, cimetidine, antacids or following total body irradiation for bone marrow transplantation, no clinically significant impairment of fluconazole absorption occurs. Physicians should be aware that drug-drug interaction studies with other medications have not been conducted, but such interactions may occur.

PREGNANCY AND LACTATION

Use during pregnancy

There have been reports of spontaneous abortion and congenital abnormalities in infants whose mothers were treated with 150mg of fluconazole as a single or repeated dose in the first trimester.

Use in pregnancy should be avoided except in patients with severe or potentially life-threatening fungal infections in whom MEDOFLUCON may be used if the anticipated benefit outweighs the possible risk to the fetus. If this drug is used during pregnancy, or if the patient becomes pregnant while taking the drug, the patient should be informed of the potential hazard to the fetus.

Effective contraceptive measure should be considered in women of child-bearing potential and should continue throughout the treatment period and for approximately 1 week (5 to 6 half-lives) after the final dose.

There have been reports of multiple congenital abnormalities in infants whose mothers were treated with high dose (400mg/day to 800mg/day) fluconazole therapy for coccidioidomycosis (an unapproved indication). The relationship between fluconazole use and these events is unclear. Adverse fetal effects have been seen in animals only at high-dose levels associated with maternal toxicity. There were no fetal effects at 5mg/kg or 10mg/kg; increases in fetal anatomical variants (supernumerary ribs, renal pelvis dilation) and delays in ossification were observed at 25mg/kg and 50mg/kg and higher doses. At doses ranging from 80mg/kg (approximately 20-60 times the recommended human dose) to 320mg/kg, embryoletality in rats were increased and fetal abnormalities included wavy rib, cleft palate and abnormal craniofacial ossification.

Case reports describe a distinctive and a rare pattern of birth defects among infants whose mothers received high dose (400-800mg/day) fluconazole during most or all of the first trimester of pregnancy. The features seen in these infants include brachycephaly, abnormal facies, abnormal calvarial development, cleft palate, femoral bowing, thin ribs and long bones, arthrogryposis and congenital heart disease.

Use during lactation

Fluconazole is found in human breast milk at concentrations similar to plasma. Breast feeding may be maintained after a single dose of 150mg fluconazole. Breast feeding is not recommended after repeated use or after high dose fluconazole.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

“MEDOFLUCON” is unlikely to have any effect on the ability to drive or use machinery

UNDESIRABLE EFFECTS

Fluconazole is generally well tolerated. The most common side effects associated with fluconazole are symptoms related to the gastrointestinal tract. These include nausea, abdominal pain, diarrhea and flatulence. After gastrointestinal symptoms, the second most commonly observed side effect was rash. Headache has been associated with fluconazole. In some patients, particularly those with serious underlying diseases such as AIDS and cancer, changes in renal and hematological function test results and hepatic abnormalities (see warning and precaution) have been observed treatment with fluconazole and comparative agents, but the clinical significance and relationship to treatment is uncertain. Exfoliative skin disorders (see warning and precautions), seizures, leukopenia including neutropenia and agranulocytosis, thrombocytopenia, and alopecia have occurred under conditions where a causal association is uncertain.

If a rash, which is considered attributable to fluconazole, develops in a patient treated for superficial fungal infection, further therapy with this agent should be discontinued. If patients with invasive/systemic fungal infections develop rashes, they should be monitored closely and fluconazole discontinued if bullous lesions or erythema multiforme develop. In rare cases, as with other azoles, anaphylaxis has been reported.

OVERDOSE

Treatment of overdosage is essentially supportive and symptomatic. Gastric lavage may be useful. Fluconazole is eliminated mainly in the urine, thus forced volume diuresis should increase elimination rate. Haemodialysis will also eliminate some fluconazole.

PHARMACOLOGY

Pharmacodynamic Properties Fluconazole, a member of new class of triazole antifungal agents, is a potent and specific inhibitor of fungal sterol synthesis. Both orally and intravenously administered fluconazole was active in a variety of animal fungal infections models. Activity has been demonstrated against opportunistic mycoses, such as infections with *Candida* spp, including systemic candidiasis in immunocompromised animals; with *Cryptococcus neoformans*, including intracranial infections; with *Microsporum* spp; and with *Trichophyton* spp. Fluconazole has also been shown to be active in animals models of endemic mycosis, including infections with *Blastomyces dermatitidis*; with *Coccidioides immitis*, including intracranial infection; and with *Histoplasma capsulatum* in normal and immunosuppressed animals. Fluconazole is highly specific for fungal cytochrome P-450 dependent enzymes. Fluconazole 50 mg daily given up to 28 days has been shown not to affect testosterone plasma concentrations in males or steroid concentrations in females of child-bearing age. Fluconazole 200-400 mg daily has no clinically significant effect on endogenous steroid levels or on ACTH stimulated response in healthy male volunteers. Interaction studies with antipyrine indicate that single or multiple doses of fluconazole 50 mg do not affect its metabolism.

Pharmacokinetic Properties The pharmacokinetics properties of fluconazole are similar following administration by the intravenous or oral route. After oral administration fluconazole is well absorbed, and plasma levels (and systemic bioavailability) are over 90% of the levels achieved after intravenous administration. Oral absorption is not affected by concomitant food intake. Peak plasma concentrations in the fasting state occur between 0.5 and 1.5 hours post-dose with a plasma elimination half life of approximately 30 hours. Plasma concentrations are proportional to dose. Ninety percent steady state levels are reached by day 4-5 with multiple once daily dosing.

Administration of loading dose (on day1) of twice the usual daily dose enables plasma levels to approximate to 90% steady-state levels by day 2. The apparent volume of distribution approximates to total body water. Plasma protein binding is low (11-12%).

Fluconazole achieves good penetration in all body fluids studied. The levels of fluconazole in saliva and sputum are similar to plasma levels. In patients with fungal meningitis, fluconazole levels in the CSF are approximately 80% the corresponding plasma levels.

High skin concentrations of fluconazole, above serum concentrations, are achieved in the stratum corneum, epidermis-dermis and eccrine sweat.

Fluconazole accumulates in the stratum corneum. At a dose of 50 mg once daily, the concentration of fluconazole after 12 days was 73 ug /g and 7 days after cessation of treatment the concentration was still 5.8ug/g. At the 150mg once-a-week dose, the concentrations of fluconazole in stratum corneum on day 7 was 23.4ug/g and 7 days after the second dose was still 7.1 ug/g.

Concentrations of fluconazole in nails after 4 months of 150 mg once-a-week dosing was 4.05 ug/g in healthy and 1.8 ug/g in diseased nails; and fluconazole was still measurable in nail samples 6 months after the end of therapy.

The major route of excretion is renal, with approximately 80% of the administered dose appearing in the urine as unchanged drug. Fluconazole clearance is proportional to creatinine clearance. There is no evidence of circulating metabolites.

The long plasma elimination half-life provides the basis for single dose therapy for vaginal candidiasis, once daily and once weekly dosing for other indications.

A study compared the saliva and plasma concentrations of a single fluconazole 100 mg dose administered in a capsule or in an oral suspension by rinsing and retaining in mouth for 2 minutes and swallowing. The maximum concentration of fluconazole in saliva after the suspension was observed 5 minutes after ingestion, and was 182 times higher than the maximum saliva concentration after the capsule which occurred 4 hours after ingestion. After about 4 hours, the saliva concentrations of fluconazole were similar. The mean AUC (0-96) in saliva was significantly greater after the suspension compared to the capsule. There was no significant difference in the elimination rate from saliva or the plasma pharmacokinetics parameters for the two formulations.

PHARMACEUTICAL PARTICULARS

Product description:

MEDOFLUCON 150mg capsules are powder Blue – powder Blue size “1” hard gelatin capsules filled with white powder.

List of Excipients: Fluconazole capsules also contain lactose, starch, silicon dioxide, sodium lauryl sulphate, magnesium stearate, gelatin and titanium dioxide (all E.P). Colourants in capsule shells are indigo carmine, E132.

Incompatibilities: None known.

Shelf Life: 48 months

Special Precautions for Storage: Store below 30°C in the original package, protected from light

Nature and Contents of Container: Medoflucon 150mg capsules: PVC/Alu blisters containing 1 capsule.
Pack size: Blistersx1.

NAME AND ADDRESS OF MANUFACTURER

Medochemie Ltd (Central Factory),
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NAME AND ADDRESS OF DISTRIBUTOR IN MALAYSIA

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DATE OF REVISION

July 2025